A August 1, 2026 — Ionic Bonding

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[10 pts]

Q1. Gravity versus Electricity

Two electrons have $q_1 = q_2 = -e$ and mass m_e; they attract gravitationally and repel electrically, with magnitudes

$$|F_E| = k\,\frac{e^2}{r^2}, \qquad |F_G| = G\,\frac{m_e^2}{r^2}.$$

Both forces fall off as r^{-2}, so their ratio is independent of the separation:

$$\frac{|F_G|}{|F_E|} = \frac{G\,m_e^2}{k\,e^2}.$$

Substituting $G = 6.674 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$, $m_e = 9.109 \times 10^{-31}$ kg, $k = 8.988 \times 10^9\ \mathrm{N\,m^2\,C^{-2}}$, and $e = 1.602 \times 10^{-19}$ C gives

$$\frac{|F_G|}{|F_E|} = 2.40 \times 10^{-43}, \qquad \log_{10}\left(\frac{|F_G|}{|F_E|}\right) \approx -42.6.$$

Gravity is about 43 orders of magnitude weaker than the electric force at every separation, so it is utterly negligible at the atomic scale. This is why Max drops it.

Q2. Potential Energy

The potential energy is the negative integral of the force, $U(r) = -\int F(r)\,\mathrm{d}r$. For the Coulomb term,

$$U_E(r) = -\int k\,\frac{q_1 q_2}{r^2}\,\mathrm{d}r = k\,\frac{q_1 q_2}{r} + C_E,$$

and for the resistance term,

$$U_R(r) = -\int \frac{nA}{r^{n+1}}\,\mathrm{d}r = \frac{A}{r^n} + C_R,$$

where the second step uses $\frac{\mathrm{d}}{\mathrm{d}r}(r^{-n}) = -n\,r^{-n-1}$, so that $-\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{A}{r^n}\right) = \frac{nA}{r^{n+1}} = F_R$. The reference $U(r \to \infty) = 0$ forces $C_E = C_R = 0$, giving

$$\boxed{U(r) = k\,\frac{q_1 q_2}{r} + \frac{A}{r^n}}.$$

The masses m_1, m_2 do not appear: the interaction potential depends only on the charges and the separation, never on the masses.

Q3. Stable Bond

A stable bond is a minimum of U, so $\frac{\mathrm{d}U}{\mathrm{d}r} = 0$. Differentiating,

$$\frac{\mathrm{d}U}{\mathrm{d}r} = -k\,\frac{q_1 q_2}{r^2} - \frac{nA}{r^{n+1}} = k\,\frac{|q_1 q_2|}{r^2} - \frac{nA}{r^{n+1}},$$

where the last equality uses $q_1 q_2 = -|q_1 q_2|$ (opposite charges). Setting this to zero,

$$k\,\frac{|q_1 q_2|}{r_0^2} = \frac{nA}{r_0^{n+1}} \implies r_0^{n-1} = \frac{nA}{k\,|q_1 q_2|},$$

so the equilibrium separation is

$$\boxed{r_0 = \left(\frac{nA}{k\,|q_1 q_2|}\right)^{1/(n-1)}}.$$

The binding energy is then

$$U_0 = U(r_0) = -k\,\frac{|q_1 q_2|}{r_0} + \frac{A}{r_0^n}.$$

Eliminating A with the equilibrium condition, $\frac{A}{r_0^n} = \frac{k\,|q_1 q_2|}{n\,r_0}$, gives

$$\boxed{U_0 = -k\,\frac{|q_1 q_2|}{r_0}\left(1 - \frac{1}{n}\right) = -\frac{n-1}{n}\,\frac{k\,|q_1 q_2|}{r_0} < 0}.$$

The negative sign is precisely what makes the bond stable.

Uniqueness. The stationarity condition rearranges to

$$r^{\,n-1} = \frac{nA}{k\,|q_1 q_2|}.$$

Because $n > 1$, the map $r \mapsto r^{\,n-1}$ is strictly increasing on $(0, \infty)$, so this equation has *at most one* positive solution; and since the right-hand side is positive while $r^{\,n-1}$ runs over all of $(0, \infty)$, it has *exactly one*. Hence U has a unique critical point r_0. To confirm it is a minimum,

$$\frac{\mathrm{d}^2 U}{\mathrm{d}r^2} = 2k\,\frac{q_1 q_2}{r^3} + \frac{n(n+1)A}{r^{n+2}} = -\frac{2k\,|q_1 q_2|}{r^3} + \frac{n(n+1)A}{r^{n+2}}.$$

At r_0, using $nA = k\,|q_1 q_2|\,r_0^{n-1}$,

$$\left.\frac{\mathrm{d}^2 U}{\mathrm{d}r^2}\right|_{r_0} = -\frac{2k\,|q_1 q_2|}{r_0^3} + \frac{(n+1)k\,|q_1 q_2|}{r_0^3} = \frac{(n-1)\,k\,|q_1 q_2|}{r_0^3} > 0.$$

Thus r_0 is a local minimum, and since it is the unique critical point it is in fact the unique global minimum.